

4a	progressive waves transfer/propagate energy and energy in a stationary waves is localised/stored, <i>OR</i> <u>amplitude</u> constant for progressive wave and varies (from max/antinode to min/zero/node) for stationary wave, <i>OR</i> for stationary waves, particles within a loop/segment are in phase and for progressive wave particles within a wavelength are out of phase Any 2 of the above.	B2
4bi	wave/microwave from source/S <u>reflects</u> at reflector R reflected and (further) incident waves <u>overlap/meet/superpose</u> waves have same <u>frequency/wavelength/period/speed</u> and (amplitude), (so stationary waves are formed)	B1 B1 B1
4bii	detector/D is moved between reflector/R and source/S maximum and minimum/zero observed on <u>meter/readings/measurements/recordings</u>	B1 B1
4biii	<u>determine/measure</u> the distance between adjacent nodes/minima or maxima/antinodes or across <u>specific number</u> of nodes/antinodes wavelength is twice distance between <u>adjacent</u> nodes/minima or maxima/antinodes (or other correct method of calculation of wavelength from measurement)	B1 B1
4c	Waves from the two sources travel the same distance to the detector so the <u>path difference is zero</u>, <u>constructive interference</u> happens so the reading is <u>maximum</u>.	M1 A1

5ai	Since the field strength is zero at a point between the spheres / there are two sections	M1
-----	--	----